

YLIVIESKA, Finland

WMO: 027550

Lat: 64.050N

Lon: 24.725E

Elev: 249

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -18.7	(c) -12.3	(d) -24.1	(e) 1.5	(f) -18.3	(g) -17.0	(h) 2.2	(i) -12.2	(j) 20.6	(k) 25.5	(l) 18.4	(m) 25.9	(n) 2.2	(o) 130	(p) 0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	80.2	65.3	76.4	62.8	73.0	61.3	67.7	76.5	65.4	73.3	63.2	70.2	7.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	93.0	71.4	62.4	85.1	68.5	60.1	78.4	66.6	32.2	76.6	30.4	73.1	28.8	69.9	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	15.5	13.8	-22.5	83.5	7.2	4.5	-27.7	86.7	-32.0	89.4	-36.0	91.9	-41.3	95.2
			-22.2	69.9	6.9	3.8	-27.1	72.6	-31.1	74.9	-35.0	77.0	-40.0	79.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	16.3	16.7	23.7	35.5	46.7	53.7	59.9	56.6	48.5	37.3	30.3	24.0	
	DBStd	17.43	13.42	13.86	10.44	6.77	7.88	6.57	6.23	6.24	6.38	7.42	8.77	11.33	
	HDD50	5336	1045	933	817	436	158	34	4	15	102	395	590	807	
	HDD65	10058	1510	1353	1282	885	568	344	183	268	496	857	1040	1272	
	CDD50	792	0	0	0	1	57	145	310	220	56	3	0	0	
	CDD65	38	0	0	0	0	1	5	24	8	0	0	0	0	
	CDH74	597	0	0	0	0	52	101	341	103	0	0	0	0	
	CDH80	112	0	0	0	0	4	21	76	11	0	0	0	0	
Wind	WSAvg	6.5	6.1	6.4	6.8	6.7	7.1	7.0	5.9	5.8	6.3	6.5	6.7	6.9	
Precipitation	PrecAvg	22.1	1.5	1.2	0.9	1.2	1.9	2.1	2.8	2.8	2.1	2.0	2.0	1.7	
	PrecMax	27.3	2.9	2.4	2.0	2.6	3.2	3.9	4.8	5.3	3.6	3.5	3.5	3.8	
	PrecMin	17.4	0.3	0.4	0.2	0.4	0.3	0.4	0.6	0.9	0.8	0.2	0.7	0.4	
	PrecStd	2.9	0.9	0.6	0.5	0.6	0.9	1.0	1.2	1.2	0.9	1.0	0.7	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.3	41.8	48.5	64.7	79.2	83.4	86.3	81.4	69.6	57.4	49.7	42.3	
		MCWB	36.7	37.9	40.8	50.6	59.0	65.4	67.4	68.2	59.7	53.0	47.7	41.4	
	2%	DB	35.9	38.5	43.5	57.5	74.4	76.5	81.8	76.9	66.1	52.9	46.5	39.9	
		MCWB	35.0	35.8	37.8	46.2	58.0	61.2	67.0	64.1	58.7	50.1	44.7	38.6	
	5%	DB	34.2	35.0	39.6	52.5	68.6	71.9	77.4	73.1	62.3	50.1	43.4	38.1	
		MCWB	33.3	33.6	35.3	43.4	55.0	59.2	64.6	62.3	56.0	48.0	42.4	36.8	
	10%	DB	32.6	33.0	36.8	48.3	62.8	68.1	74.1	69.7	59.4	47.8	40.8	35.9	
		MCWB	31.9	32.0	33.7	40.9	51.3	57.2	63.2	60.7	54.7	45.9	39.9	34.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	38.5	41.7	51.8	62.2	67.2	71.6	70.1	62.1	53.7	47.8	41.5
			MCDB	38.0	41.2	48.0	62.9	74.6	78.0	79.8	78.4	66.2	56.0	49.3	41.8
2%		WB	35.3	36.0	38.3	47.6	59.1	64.3	68.7	66.7	59.4	50.4	44.7	39.0	
		MCDB	35.8	38.1	42.2	56.3	71.0	73.8	77.7	73.7	64.6	52.2	46.0	39.6	
5%		WB	33.5	33.9	36.1	43.9	56.1	61.1	66.6	63.9	57.2	48.2	42.3	36.9	
		MCDB	34.1	35.0	38.9	51.0	67.6	69.4	74.9	69.7	61.4	49.9	43.2	37.9	
10%		WB	32.1	32.2	34.1	41.3	52.7	58.4	64.4	61.8	55.4	45.8	40.0	34.9	
		MCDB	32.7	32.9	36.1	47.2	61.0	65.9	72.4	67.1	58.8	47.3	40.7	35.7	
Mean Daily Temperature Range		5% DB	MDBR	11.8	13.9	17.4	18.4	21.5	20.7	21.0	20.0	16.8	11.1	7.9	9.6
			MCDBR	7.6	9.6	16.7	27.2	32.4	29.1	27.4	26.7	22.1	12.6	8.8	8.0
	MCWBR		7.2	8.3	12.4	17.2	17.3	16.1	14.3	15.1	14.7	10.1	8.2	7.7	
	5% WB	MCDBR	7.4	8.8	14.9	24.9	30.2	23.8	23.9	20.8	19.1	11.8	8.6	7.6	
		MCWBR	7.2	7.9	11.4	16.6	16.8	14.1	13.0	12.9	13.2	10.0	8.1	7.4	
Clear-Sky Solar Irradiance	taub	0.217	0.281	0.310	0.350	0.346	0.349	0.369	0.353	0.329	0.308	0.357	0.355		
	taud	1.987	2.140	2.199	2.308	2.443	2.460	2.422	2.460	2.514	2.375	2.620	2.623		
	Ebn at Noon	111	197	244	261	275	276	266	260	243	192	103	55		
	Edh at Noon	9	19	28	32	30	31	31	28	21	16	8	5		
All-Sky Solar Radiation	RadAvg	46	220	688	1187	1552	1714	1593	1202	681	255	60	17		
	RadStd	5	41	68	110	165	148	156	121	72	45	11	2		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page